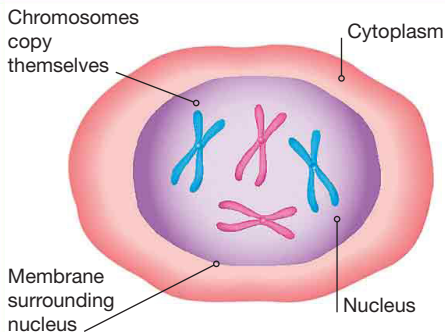
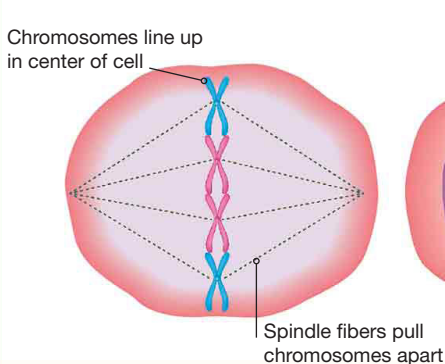


Dividing cells

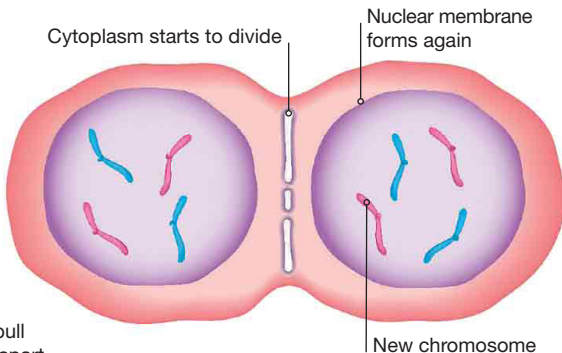
We all start life as a single cell. That cell divides again and again to produce the trillions of cells needed to build a body. Without cell division—or mitosis—the body would be unable to grow. It would also be unable to repair itself by replacing worn-out, damaged, or lost cells.



Single cell



Lining up



Separation

Identical offspring

In mitosis, a cell divides to produce two identical cells. Inside its nucleus, chromosomes hold the instructions to build and run the cell. First, each chromosome copies itself. Then, the two-stranded chromosomes line up. Next, the two strands are pulled apart to opposite ends of the cell. Finally, the cytoplasm divides to form two new, identical cells.